

Mentora

Smart Learning, Bright future

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MVP Scoping & Workplan

Project Overview

The goal of this project is to design and develop an AI-based private tutoring application for children and teenagers aged 8–18. The application aims to provide personalized academic support by combining artificial intelligence with interactive learning materials such as explanations, quizzes, videos, presentations, and educational games.

Unlike traditional learning platforms that mainly provide static content, this system focuses on active interaction between the student and an AI tutor. The tutor adapts explanations, difficulty level, and practice content based on the student's responses and learning progress. The project emphasizes accessibility, personalization, and engagement across a wide age range.

MVP Scope Definition

The Minimum Viable Product (MVP) is designed to demonstrate the core value of the system while keeping the scope realistic for a final project timeline. The MVP focuses on one primary user type: the student.

In-Scope Features

The MVP will include the following core functionalities:

1. **User Profile and Authentication**

Students will be able to log into the system and create a basic profile. The profile will store age group, selected subjects, and learning progress. This information will be used to personalize explanations and practice questions.

2. **Subject and Topic Selection**

The system will allow students to select a subject (e.g., mathematics or English) and a specific topic. The initial version will focus on a limited number of predefined topics to ensure quality and correctness.

3. **AI-Based Tutoring Interaction**

The AI tutor will:

- Explain learning material in clear, age-appropriate language
- Answer student questions related to the selected topic

- Adjust the explanation style and difficulty based on student performance
 - Actively guide the learning process by asking questions, reinforcing understanding, and connecting current topics to previously learned material
4. **Interactive Practice and Feedback**
After explanations, students will receive short quizzes or practice questions. The AI tutor will provide immediate feedback, explain mistakes, and suggest additional practice when needed.
 5. **Progress Tracking**
The system will store quiz results and basic interaction history, allowing the tutor to adapt future explanations and difficulty levels. In addition, the system will provide the student with a clear and accessible analysis of their learning progress, including performance trends over time, strengths and weaknesses across topics, and indicators of improvement. This analysis is intended to help students better understand their own learning process and take a more active role in monitoring their academic progress.
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Out-of-Scope Features (Future Work)

To maintain a focused MVP, the following features are explicitly excluded from the current implementation:

- Parent or teacher dashboards
- Voice-based interaction
- Advanced gamification (badges, leaderboards, avatars)
- Multi-language support
- Integration with school systems or external learning platforms

These features may be discussed as future extensions but will not be implemented in the MVP.

Workplan and Timeline

The development process is divided into structured phases to ensure gradual progress and risk reduction.

Phase 1: Requirements Analysis and Concept Refinement

This phase focuses on defining the user flow, functional requirements, and system boundaries. The outcome will be a clear description of what the system should do and what it will not do.

Deliverables:

- Requirements document
- User flow diagrams

Phase 2: Content and Learning Structure Design

In this phase, the structure of learning content will be defined, including how explanations, quizzes, and feedback are organized. This step is essential to ensure that the AI tutor produces controlled and educationally appropriate responses.

Deliverables:

- Topic structure
- Question and feedback templates

Phase 3: AI Tutor Integration

This phase includes integrating a pre-trained language model to serve as the AI tutor. Prompt design and response constraints will be implemented to ensure clarity, correctness, and age-appropriate language.

Deliverables:

- AI tutor module
- Prompting logic

Phase 4: Backend Development

The backend will handle user management, learning sessions, and progress tracking. APIs will be implemented to connect the frontend with the AI tutor and the database.

Deliverables:

- Backend services
- Database schema

Phase 5: Frontend Development

A simple and intuitive user interface will be developed, allowing students to interact with the tutor, answer questions, and view feedback.

Deliverables:

- User interface prototype
 - Integrated frontend application
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Phase 6: Testing and Evaluation

The system will be tested for functionality, usability, and logical consistency. Special attention will be given to response quality and user experience.

Deliverables:

- Test cases
 - Evaluation report
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Technical Feasibility Review

Feasibility Analysis

The proposed solution is technically feasible using existing artificial intelligence technologies and standard web development tools. The project does not require training machine learning models from scratch, which significantly reduces complexity and development time.

Large Language Models (LLMs) can generate natural language explanations, answer questions, and adapt responses based on context. By combining these models with structured prompts and predefined content, the system can deliver reliable and educationally appropriate tutoring experiences.

Technologies and Tools

- **Artificial Intelligence:**
Pre-trained Large Language Models accessed via API, used for explanation generation and interactive tutoring.
 - **Backend:**
Python-based server (e.g., Flask or FastAPI) for handling application logic and AI integration.
 - **Frontend:**
Web-based interface using standard frontend frameworks.
 - **Database:**
A database for storing user profiles, learning progress, and quiz results.
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Risks and Mitigation Strategies

One major technical risk is the possibility of incorrect or misleading explanations generated by the AI. This risk will be mitigated by constraining prompts and combining AI responses with predefined learning content.

Another risk is scope creep, which could jeopardize timely completion. This will be addressed by strictly adhering to the MVP scope.

Ethical considerations related to AI usage by children are also acknowledged. The system will avoid collecting sensitive personal data and will focus exclusively on educational interactions.

Feasibility Conclusion

Based on current technologies and available development tools, the project is achievable within the given timeframe. The proposed architecture and limited MVP scope ensure that the system can be implemented, tested, and evaluated as a functional prototype.

High-Level Design (HLD)

System Architecture Overview

The system follows a modular client-server architecture composed of four main components:

- Frontend Application**
Responsible for user interaction, displaying learning content, and collecting student input.
 - Backend Server**
Handles authentication, session management, progress tracking, and communication with the AI module.
 - AI Tutor Module**
Processes student input and generates explanations, questions, and feedback using a language model.
 - Database**
Stores user data, learning history, and performance metrics.
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High-Level Interaction Flow

User Story 1: Learning a New Topic

Description:

The student wants to learn a new topic from scratch in a structured and guided way.

Flow:

1. The student logs into the system
2. The student selects a subject and a new topic to learn
3. The AI tutor presents a structured explanation, using interactive elements such as short explanations, examples, and educational games when relevant
4. The student asks follow-up questions during or after the explanation
5. The system presents practice quizzes or short exercises to reinforce understanding
6. The AI tutor provides immediate feedback, explains mistakes, and adjusts the difficulty level if needed
7. The system stores the student's progress and uses it to personalize future learning sessions

Goal:

To help the student build a clear and confident understanding of new material through guided explanations and active practice.

User Story 2: Solving a Specific Learning Problem**Description:**

The student arrives with a specific difficulty, such as homework they do not understand, an unclear class presentation, or a topic they missed in class.

Flow:

1. The student logs into the system
2. The student uploads homework, notes, or class presentations, or describes the specific problem in text
3. The AI tutor analyzes the provided material and identifies the relevant concepts
4. The AI tutor explains the material step by step, similar to a private tutor, adjusting the explanation based on the student's questions
5. The student asks targeted questions about specific parts they do not understand
6. The AI tutor revisits prerequisite topics when needed and connects them to the current problem
7. The system tracks the interaction and updates the student's learning progress

Goal:

To provide personalized, on-demand tutoring that helps the student fully understand specific academic difficulties, rather than only delivering generic explanations.

Design Rationale

The modular design allows clear separation of concerns and simplifies future extensions. The AI tutor is isolated from the core system logic, enabling changes to the AI model without major architectural modifications.
